

MEME16203 Linear Models**Assignment 1****UNIVERSITI TUNKU ABDUL RAHMAN**

Faculty:	FES	Unit Code:	MEME16203
Course:	MAC	Unit Title:	Linear Models
Year:	1,2	Lecturer:	Dr Yong Chin Khian
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Due by:			

Q1. Use the definition of orthogonal and idempotent matrices and properties of determinants in the lecture notes to prove the following results:

- (a) If $\mathbf{A}$ is an orthogonal matrix, then $|\mathbf{A}|$ is either 1 or -1. (Hint: use the definition of an orthogonal matrix and consider the determinant of an identity matrix.)
- (b) If $\mathbf{W}$ is an idempotent matrix, then $|\mathbf{W}|$ is either 0 or 1.
- (c) If $\mathbf{W}$ is an idempotent matrix, then the eigenvalues are either 0 or 1.

(15 marks)

Q2. For a $k \times k$ symmetric matrix $\mathbf{A}$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k$ and eigenvectors $\mathbf{u} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k]^T$ with $\mathbf{u}_i^T \mathbf{u}_i = 1$ and $\mathbf{u}_i^T \mathbf{u}_j = 0$.

- (a) Use spectral decomposition to show that $\text{trace}(\mathbf{A}) = \sum_{i=1}^k \lambda_i$.
- (b) Show that $\text{Rank}(\mathbf{A}) = \text{number of nonzero eigenvalues}$.
- (c) Show that if $\mathbf{A}$ idempotent, then $\text{Rank}(\mathbf{A}) = \text{trace}(\mathbf{A})$.

(15 marks)

Q3. You are given:

$$\mathbf{P}_{n \times n} = \begin{bmatrix} \mathbf{h}^T = \frac{1}{\sqrt{n}} \mathbf{1}_n^T \\ \mathbf{H}_0 \end{bmatrix}$$

where $\mathbf{H}_0$ = last $(n-1)$ rows of $\mathbf{H}_{n \times n}$ with $\mathbf{H}_0$ having its r th row as

$$\left[\frac{1}{\sqrt{r(r+1)}} \mathbf{1}_r^T \quad \frac{-r}{\sqrt{r(r+1)}} \mathbf{0}_{(n-r-1) \times 1} \right]$$

$$r = 1, 2, \dots, n-1.$$

- (a) Suppose $n = 5$, show that $\mathbf{P}$ is orthogonal.
- (b) Suppose $\mathbf{y}^T = [y_1, y_2, \dots, y_5]^T$, find $\mathbf{y}^T \mathbf{h}^T \mathbf{h} \mathbf{y}$ and $\mathbf{y}^T \mathbf{H}_0^T \mathbf{H}_0 \mathbf{y}$.

(10 marks)

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Q4. You are given:

$$\mathbf{P}_{n \times n} = \begin{bmatrix} \mathbf{h}^T = \frac{1}{\sqrt{n}} \mathbf{1}_n^T \\ \mathbf{H}_0 \end{bmatrix}$$

where $\mathbf{H}_0$ = last $(n - 1)$ rows of $\mathbf{H}_{n \times n}$ with $\mathbf{H}_0$ having its r th row as

$$\left[\frac{1}{\sqrt{r(r+1)}} \mathbf{1}_r^T \quad \frac{-r}{\sqrt{r(r+1)}} \mathbf{0}_{(n-r-1) \times 1} \right]$$

$$r = 1, 2, \dots, n - 1.$$

Suppose $\mathbf{y}^T = [163.5, 165.7, 153.2, 175.8, 155.2, 160.9, 169.0, 158.5, 159.7, 160.5]^T$, find $\mathbf{y}^T \mathbf{h}^T \mathbf{h} \mathbf{y}$ and $\mathbf{y}^T \mathbf{H}_0^T \mathbf{H}_0 \mathbf{y}$.

(10 marks)

Q5. Suppose $\mathbf{Z} = \mathbf{1}_{4 \times 1}$, $\mathbf{G} = 9$, $\mathbf{R} = 64 \mathbf{I}_{4 \times 4}$. If $\Sigma = \mathbf{Z} \mathbf{G} \mathbf{Z}^T + \mathbf{R}$, find Σ^{-1} .

(10 marks)

Q6. Suppose $\mathbf{X}$ is an $n \times p$ matrix. Prove that $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{P}_\mathbf{X})$.

(20 marks)

Q7. A useful result from linear algebra (that you may use it without proof) is as follows:

$$\text{rank}(\mathbf{UV}) \leq \min[\text{rank}(\mathbf{U}), \text{rank}(\mathbf{V})]$$

for any two matrices $\mathbf{U}$ and $\mathbf{V}$ with dimensions that allow multiplication (number of columns of $\mathbf{U}$ equals the number of rows of $\mathbf{V}$). In words, this result says that the rank of a product of matrices is no greater than the rank of any matrix in the product. Show that for any matrix $\mathbf{X}$, $\text{rank}(\mathbf{X}) = \text{rank}(\mathbf{P}_\mathbf{X})$.

(20 marks)